Screen Reader User Survey #10 Results

Introduction

In December 2023 and January 2024, WebAIM surveyed preferences of screen reader users. We received 1539 valid responses. This was a follow-up to 9 previous surveys that were conducted between January 2009 and June 2021.

A few disclaimers and notices:

- Totals may not equal 100% due to rounding.
- Total responses (n) for each question may not equal 1539 due to some respondents not answering a particular question.
- The sample was not controlled and may not represent all screen reader users.
- We hope to conduct additional surveys of this nature again in the future. If you have recommendations or questions that you would like us to ask, please contact us.

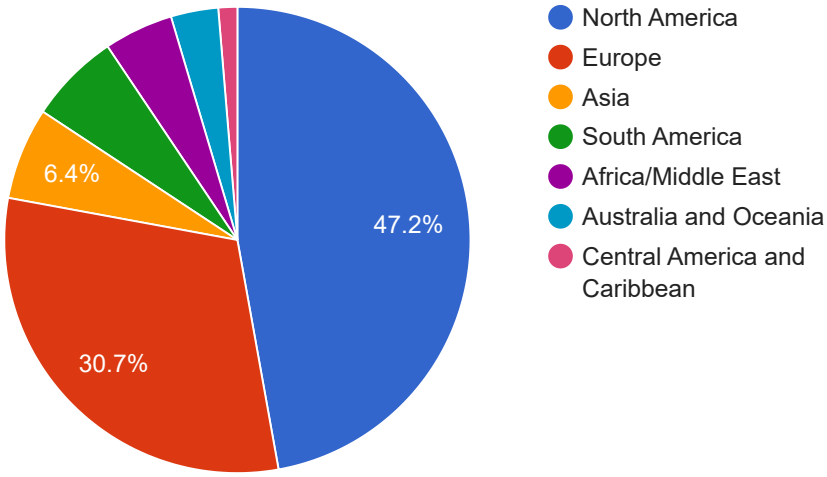
Supported by BrowserStack

Support for this research is funded in part by a donation from

BrowserStack

Demographics

Region

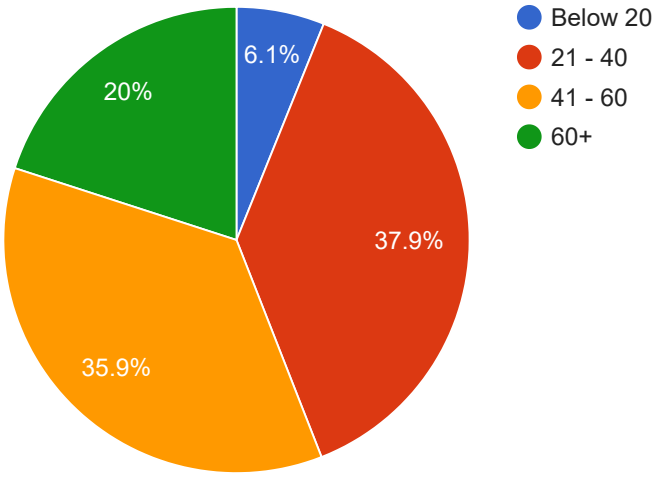


Please select your region.

Region	# of respondents	% of respondents
North America	717	47.2%
Europe	467	30.7%
Asia	97	6.4%
South America	96	6.3%
Africa/Middle East	73	4.8%
Australia and Oceania	50	3.3%
Central America and Caribbean	20	1.3%

This survey saw a majority of respondents from outside North America, thus providing better representation of the global screen reader user audience. When survey responses were notably different between regions, this is noted below.

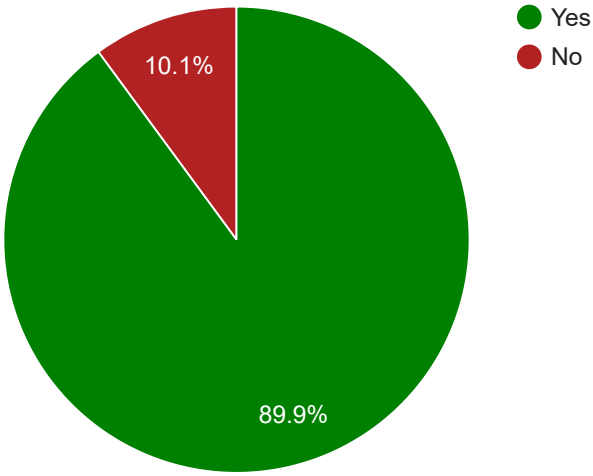
Age



Please indicate your age.

Age	# of respondents	% of respondents
Below 20	94	6.1%
21 - 40	584	37.9%
41 - 60	553	35.9%
60+	308	20.0%

Disability



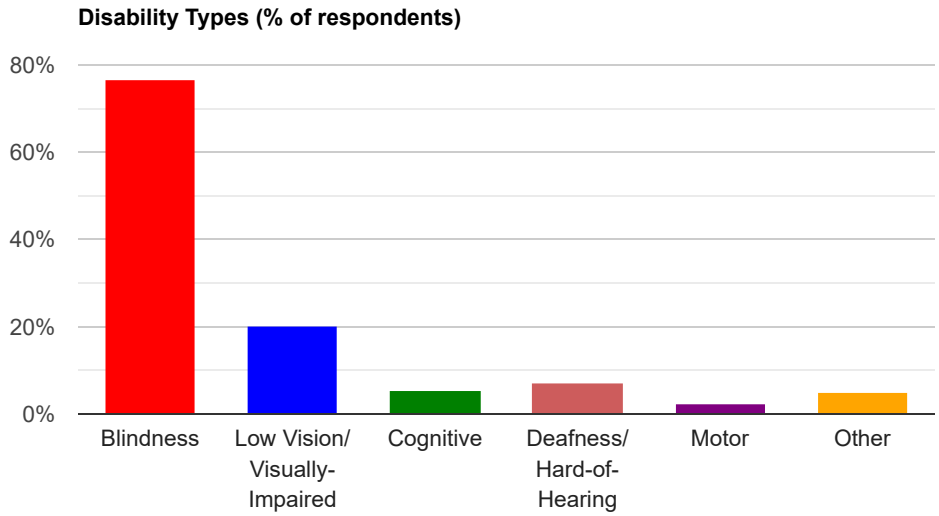
Do you use a screen reader due to a disability?

Response	# of Respondents	% of Respondents

Yes	1372	89.9%
No	154	10.1%

Responses are predominantly very similar for respondents with and without disabilities. Any notable differences are indicated below to highlight differences in practices or perceptions between disabled and non-disabled respondents.

Disability Types

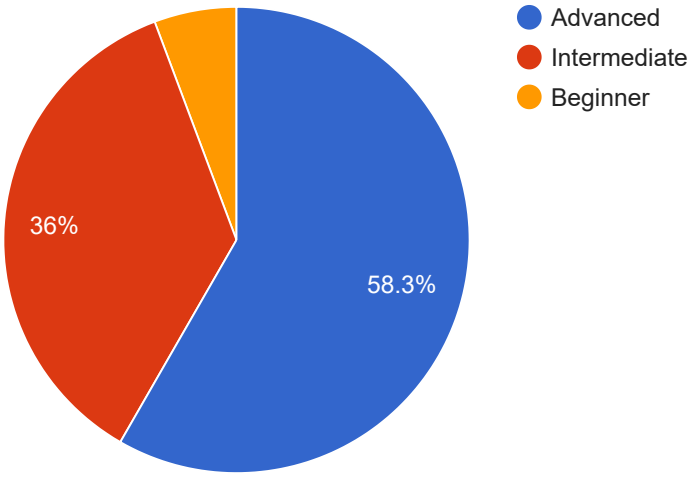


Which of the following disabilities do you have?

Response	# of respondents	% of respondents
Blindness	1179	76.6%
Low Vision/Visually-Impaired	306	19.9%
Cognitive or Learning	80	5.2%
Deafness/Hard-of-Hearing	104	6.8%
Motor	34	2.2%
Other	75	4.9%

257 respondents (16.7%) reported multiple disabilities. 81 respondents (5.3%) reported being both deaf/hard-of-hearing and blind.

Screen Reader Proficiency

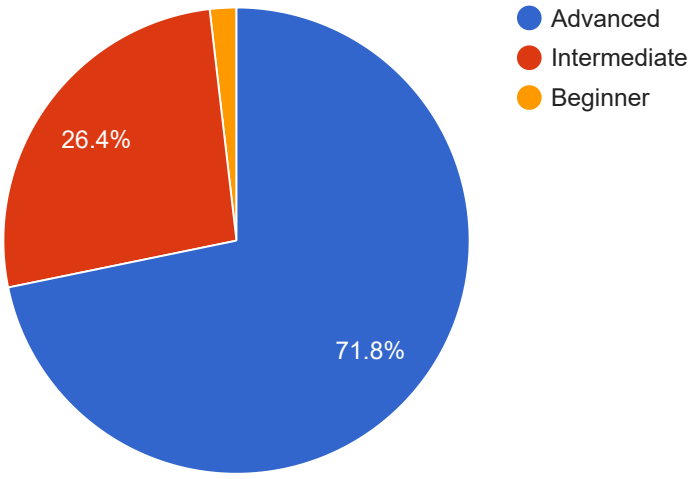


Please rate your screen reader proficiency.

Response	# of respondents	% of respondents
Advanced	888	58.3%
Intermediate	548	36.0%
Beginner	87	5.7%

Those who use screen readers due to a disability reported themselves as more proficient with screen readers—63% of those with disabilities considered their proficiency to be "Advanced" compared to only 18.2% of those without disabilities.

Internet Proficiency

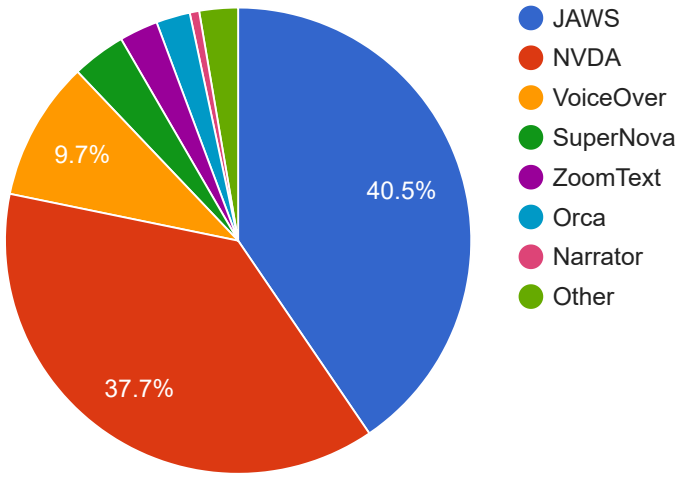


Please rate your proficiency using the Internet.

Response	# of respondents	% of respondents
Advanced	1091	71.8%
Intermediate	401	26.4%
Beginner	28	1.8%

Those without disabilities rated themselves as more proficient than those with disabilities.

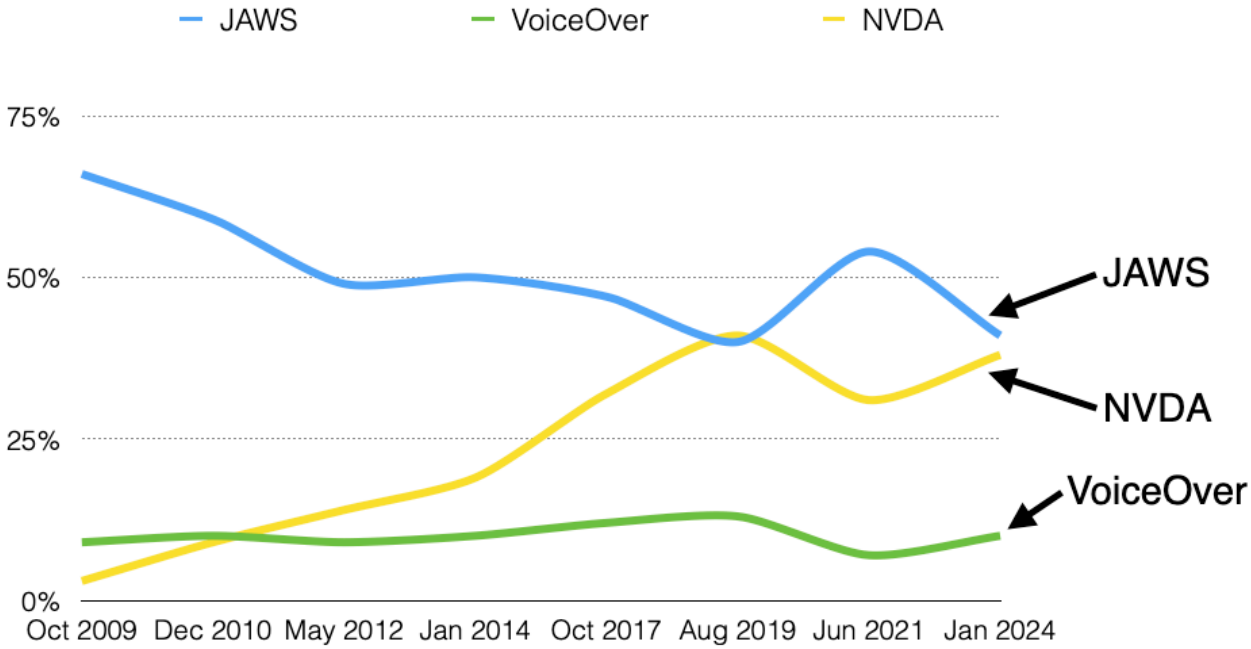
Primary Desktop/Laptop Screen Reader



Which of the following is your primary desktop/laptop screen reader?

Response	# of respondents	% of respondents
JAWS	619	40.5%
NVDA	577	37.7%
VoiceOver	148	9.7%
Dolphin SuperNova	57	3.7%
ZoomText/Fusion	41	2.7%
Orca	36	2.4%
Narrator	10	0.7%
Other	41	2.7%

The following chart shows historical trends for primary screen reader usage.



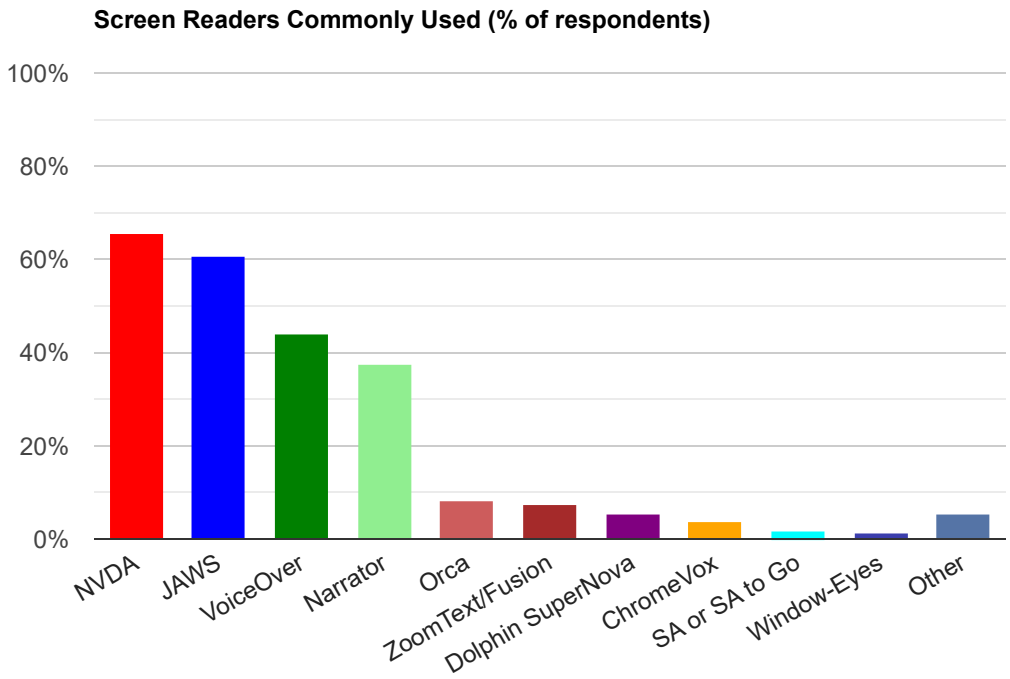
Historically JAWS usage had been in decline until 2019 when JAWS and NVDA were nearly the same. The 2021 survey saw an increase in JAWS usage and decrease in NVDA usage, but in 2024 the usage of both are again nearly the same at 41% for JAWS and 38% for NVDA.

Respondents with disabilities are more likely to use JAWS and NVDA and less likely to use VoiceOver as their primary screen reader than respondents without disabilities. 8.2% of respondents with disabilities primarily use VoiceOver (up from 5.5% in 2021), compared to 23.7% of respondents without disabilities.

Primary usage varied greatly by region. JAWS usage was higher than NVDA in North America (55.5% vs. 24.0%) and Australia (45.8% vs. 37.5%), though JAWS usage was lower than NVDA in Europe (29.7% vs. 37.2%), Africa/Middle East (23.3% vs. 69.9%), and Asia (22.9% vs. 70.8%).

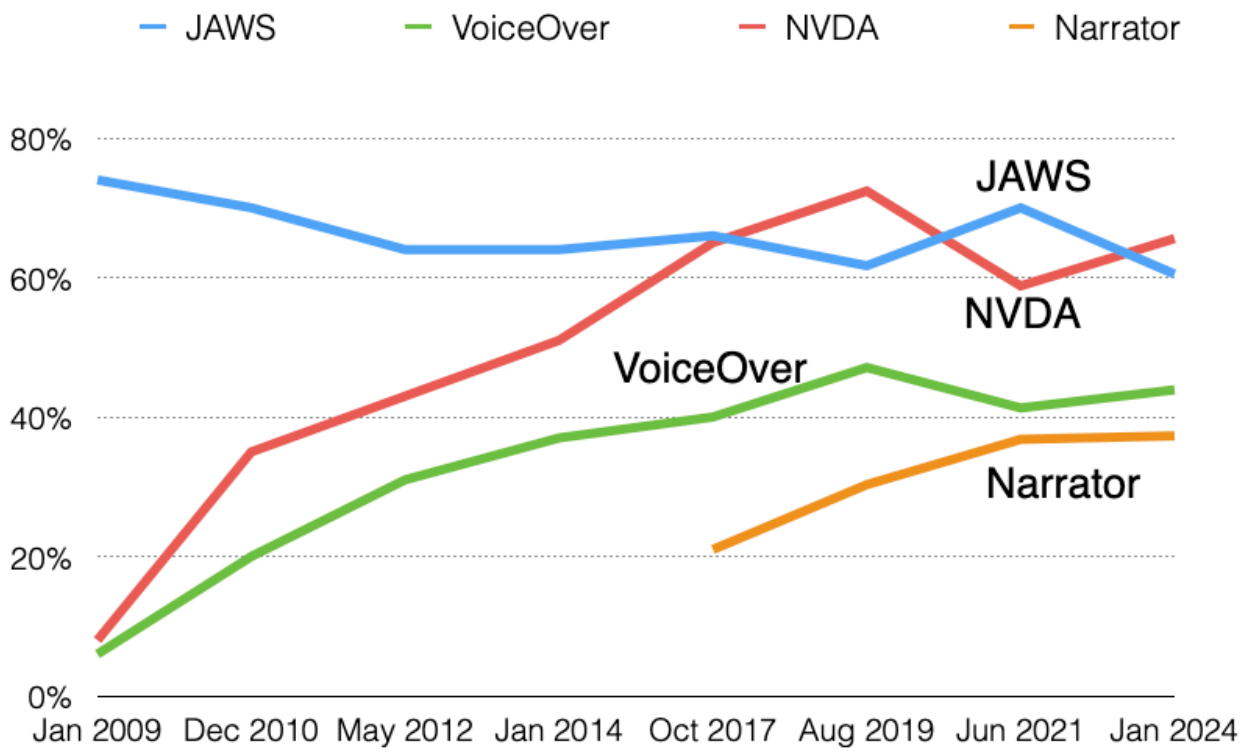
Beyond the three most popular primary screen readers, other screen readers comprised a total of 12.2% of usage.

Screen Readers Commonly Used



Which of the following desktop/laptop screen readers do you commonly use?

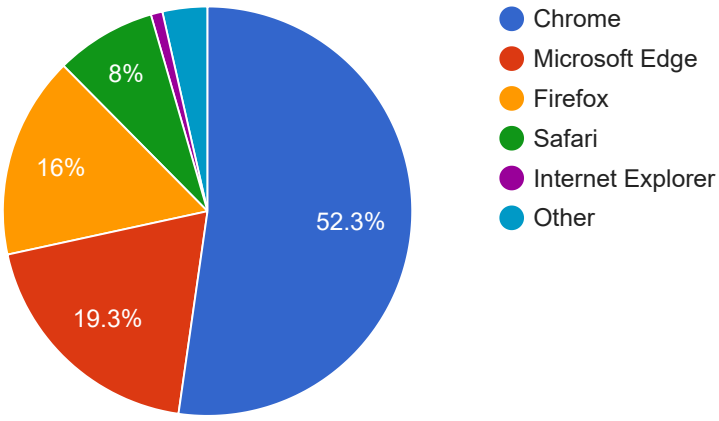
Response	# of respondents	% of respondents
NVDA	1009	65.6%
JAWS	931	60.5%
VoiceOver	675	43.9%
Narrator	574	37.3%
Orca	127	8.3%
ZoomText/Fusion	115	7.5%
Dolphin SuperNova	83	5.4%
ChromeVox	59	3.8%
System Access or System Access to Go	28	1.8%
Window-Eyes	18	1.2%
Other	79	5.1%



NVDA is again the most commonly used screen reader at 65.6% of respondents outpacing JAWS at 60.5%. Narrator—freely available in Windows for several years—is the primary screen reader of only .7% of respondents, but is commonly used by 37.3% of respondents.

71.6% of respondents use more than one desktop/laptop screen reader. 43% use three or more, and 17.4% use four or more different screen readers. VoiceOver users most commonly use additional screen readers.

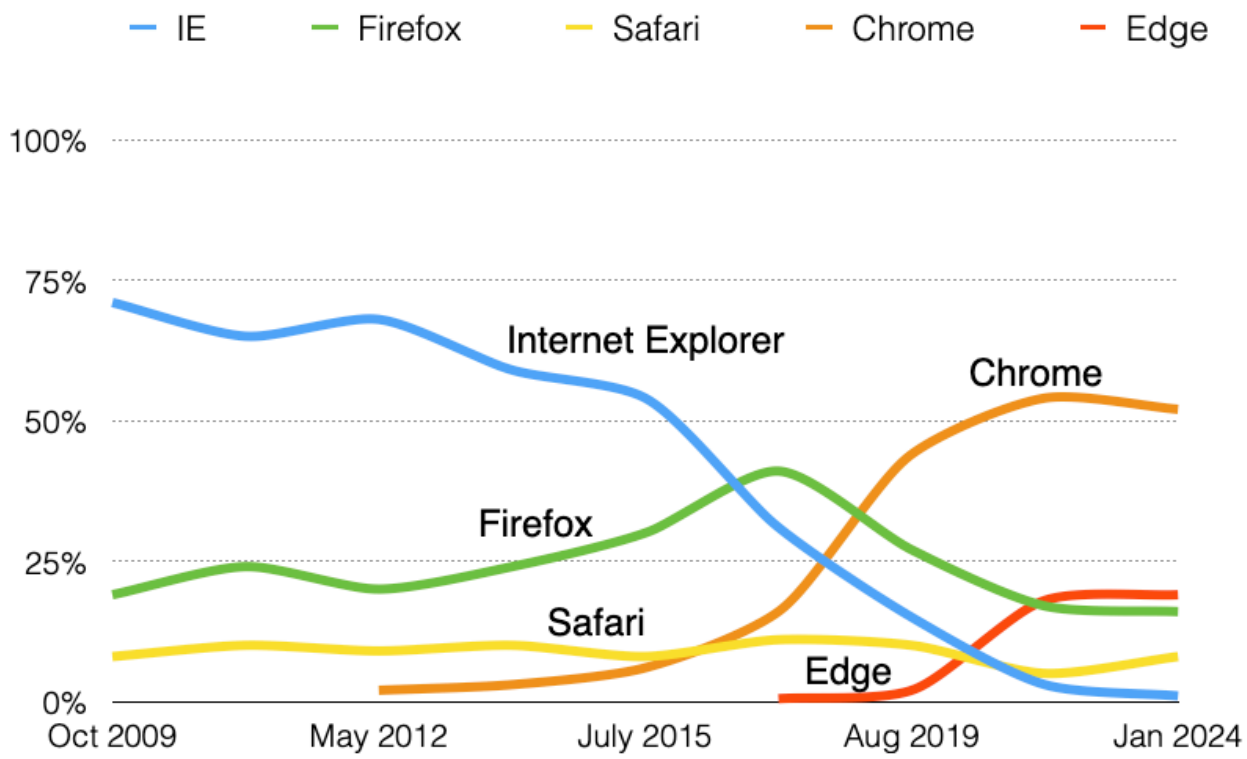
Browsers



When using your primary screen reader, which browser do you use most often?

Response	# of respondents	% of respondents
Chrome	795	52.3%
Microsoft Edge	294	19.3%

Firefox	243	16.0%
Safari	121	8.0%
Internet Explorer	14	.9%
Other	54	3.6%



Browser usage remains mostly unchanged since 2021, with Chrome and Internet Explorer seeing small decreases in usage and Safari a small increase in usage.

Screen Reader / Browser Combinations

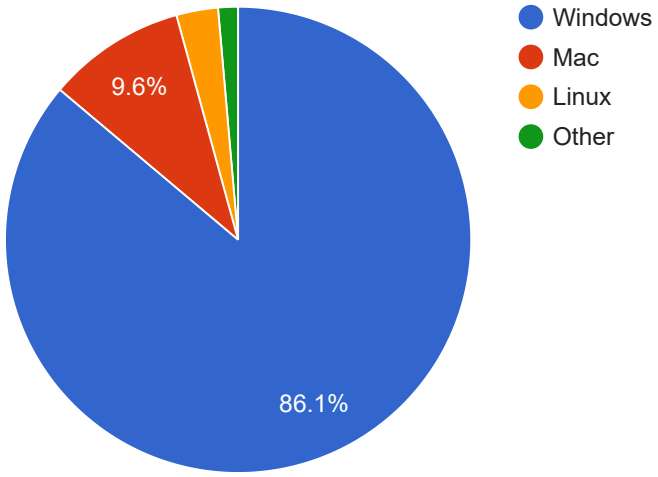
Most common screen reader and browser combinations

Screen Reader & Browser	# of Respondents	% of Respondents
JAWS with Chrome	373	24.7%
NVDA with Chrome	323	21.3%
JAWS with Edge	173	11.4%
NVDA with Firefox	152	10.0%
VoiceOver with Safari	107	7.0%
NVDA with Edge	75	5.0%
JAWS with Firefox	39	2.6%
VoiceOver with Chrome	30	2.0%
Orca with Firefox	29	1.9%
Dolphin SuperNova with Chrome	24	1.6%
ZoomText/Fusion with Chrome	18	1.2%
ZoomText/Fusion with Edge	16	1.1%

Other combinations	154	10.2%

There are **many** combinations of browsers and screen readers in use, with JAWS with Chrome the most common.

Operating System

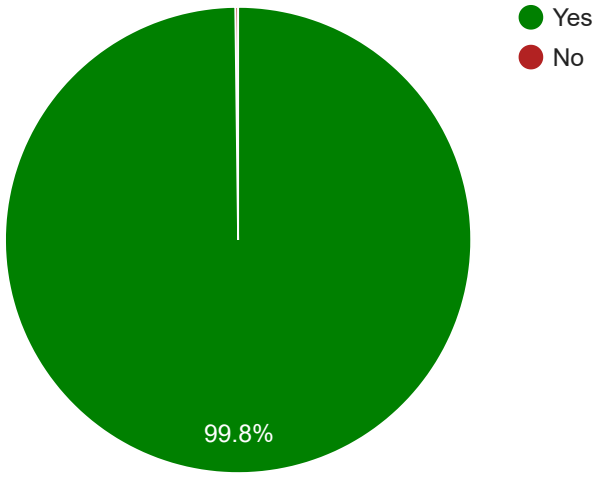


What operating system are you on when using your primary desktop/laptop screen reader?

Response	# of respondents	% of respondents
Windows	1311	86.1%
Mac	146	9.6%
Linux	44	2.9%
Other	21	1.4%

Respondents without disabilities were nearly 3 times more likely to use Mac OS than respondents with disabilities.

JavaScript

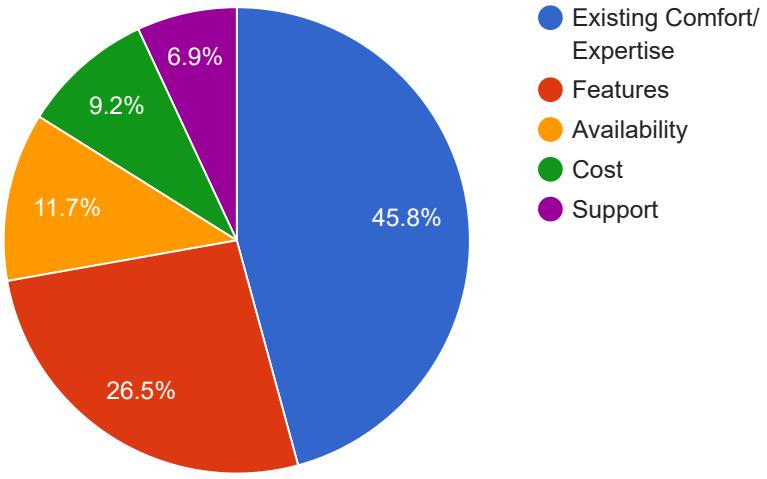


JavaScript Enabled

Response	% of Respondents
Yes	99.8%
No	.2%

JavaScript support was detected with the survey form submission. Nearly all respondents had JavaScript enabled.

Reason for Use

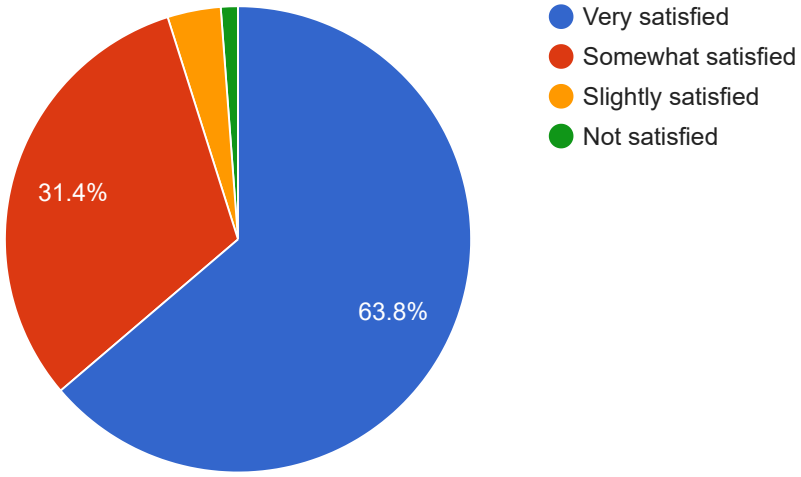


What is the main reason for using your primary screen reader?

Response	# of respondents	% of respondents
Existing Comfort/Expertise	685	45.8%
Features	396	26.5%
Availability	175	11.7%
Cost	137	9.2%
Support	104	6.9%

Existing comfort and features went down slightly, and Availability, Cost, and Support went up slightly.

Screen Reader Satisfaction



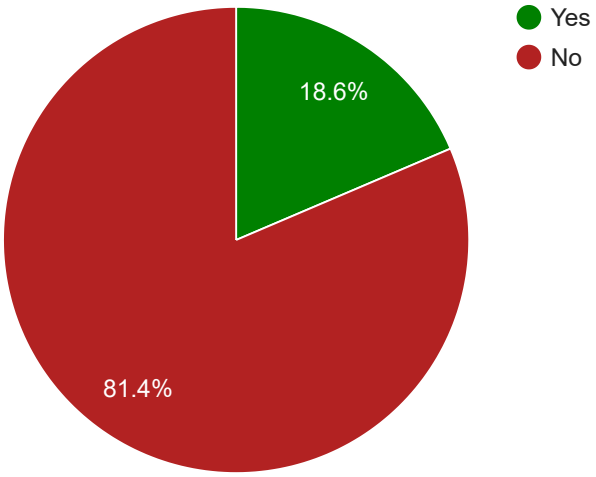
How satisfied are you with your primary screen reader?

Response	# of respondents	% of respondents
Very satisfied	966	63.8%
Somewhat satisfied	475	31.4%
Slightly satisfied	56	3.7%
Not satisfied	18	1.2%

Respondents indicating that they are very or somewhat satisfied by their primary screen reader:

- NVDA - 97.6%
- JAWS - 95.6%
- VoiceOver - 92.4%
- Narrator - 88.9%

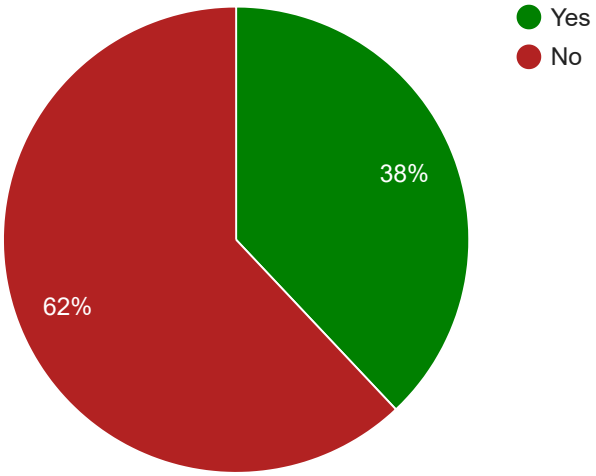
Home vs. Work



Response	# of respondents	% of respondents
Yes	266	18.6%
No	1165	81.4%

31% of VoiceOver users reported using a different screen reader at work/school vs. home, versus 19% for NVDA, and 15% for JAWS.

Braille Output



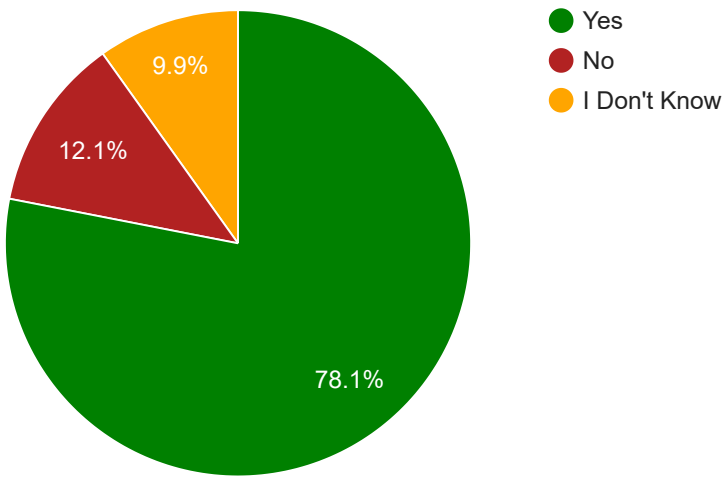
Do you use braille output with your screen reader?

Response	# of respondents	% of respondents

Yes	510	38%
No	833	62%

Because it would not generally be expected that users without disabilities would use Braille, they have been omitted from these data. Braille usage at 38% is up slightly from 33.3% in 2017 and 27.7% in 2012. 54.2% of VoiceOver users used Braille compared to 42.4% of NVDA users and 35.1% of JAWS users.

Free/Low-cost vs. Commercial



Do you see free or low-cost desktop screen readers (such as NVDA or VoiceOver) as currently being viable alternatives to more expensive commercial screen readers?

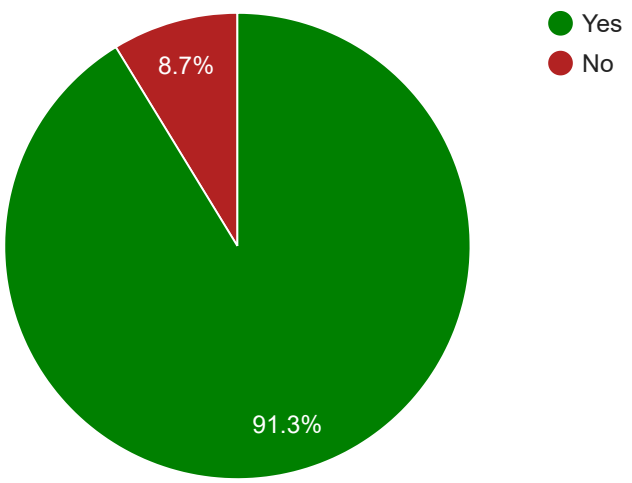
Response	# of respondents	% of respondents
Yes	1171	78.1%
No	181	12.1%
I Don't Know	148	9.9%

While the positive perception of free or low-cost screen readers has increased significantly from 48% in October 2009, responses to this question over the last several years have been largely unchanged.

Only 67% of JAWS users answered "Yes" compared to an overwhelming 91% of VoiceOver users and 94% of NVDA users. Those that use free or low-cost screen readers as their primary screen reader have a much better perception of them than those who do not. Respondents with "Advanced" screen reader proficiency were also more favorable of free/low-cost screen readers.

Mobile Screen Readers

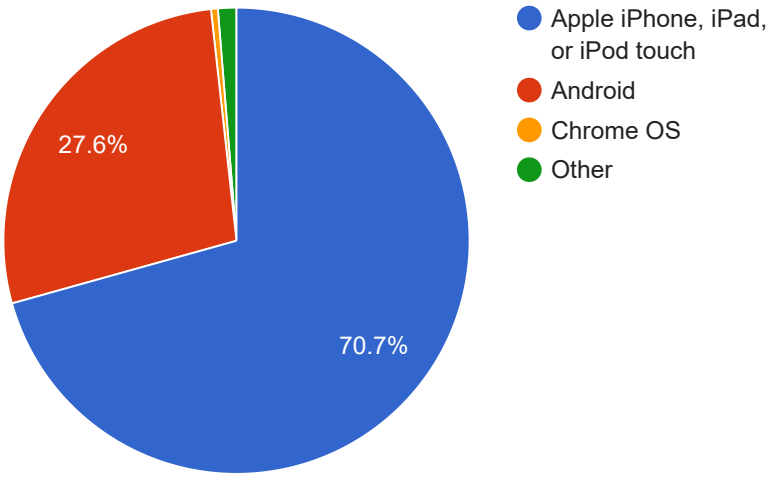
Mobile Usage



Yes	1379	91.3%
No	132	8.7%

91.3% of respondents report using a screen reader on a mobile device. Respondents with disabilities (93.6%) are more likely to use a mobile screen reader than respondents without disabilities (70.4%).

Mobile Platforms

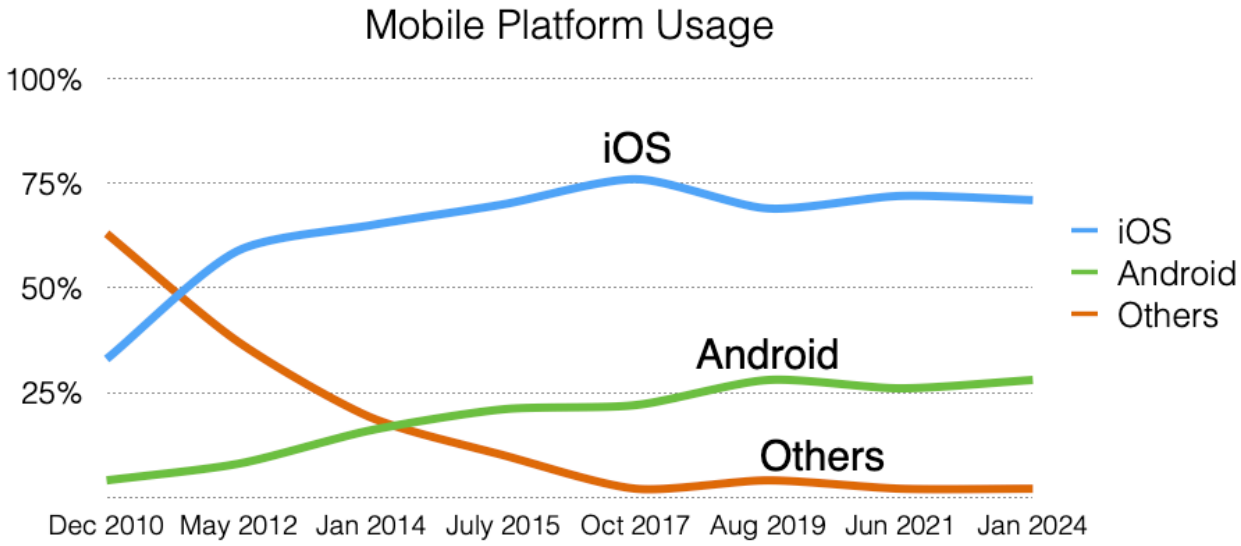


Which of the following is your primary mobile/tablet platform?

Response	# of respondents	% of respondents
Apple iPhone, iPad, or iPod touch	1048	70.6%

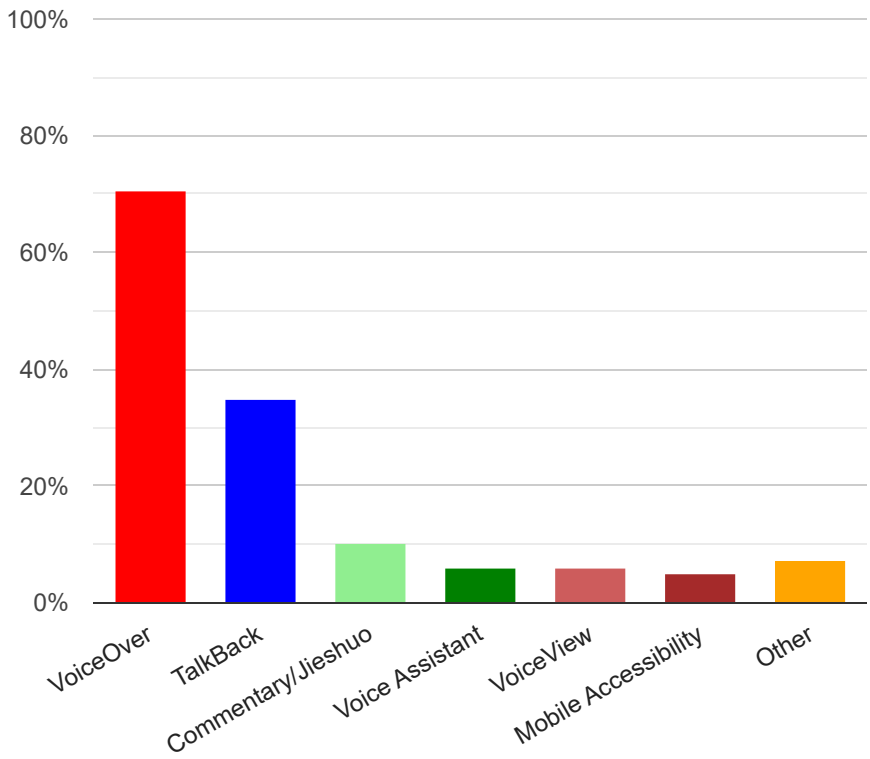
Android	409	27.6%
Chrome OS	7	0.5%
Other	19	1.3%

Respondents with disabilities (72.4%) used iOS devices at a higher rate than those without disabilities (56%). Usage of iOS devices was significantly higher in North America (84%), Australia (75%), and Europe/UK (72%) than in Asia (40%), Africa/Middle East (34%), and South America (31%). Respondents with more advanced screen reader and internet proficiency were much more likely to use iOS over Android.



iOS devices continue to dominate the mobile screen reader market. Usage of platforms other than iOS and Android (Chrome OS, Windows Phone, Nokia, etc.) combined represent only 1.8% of reported usage.

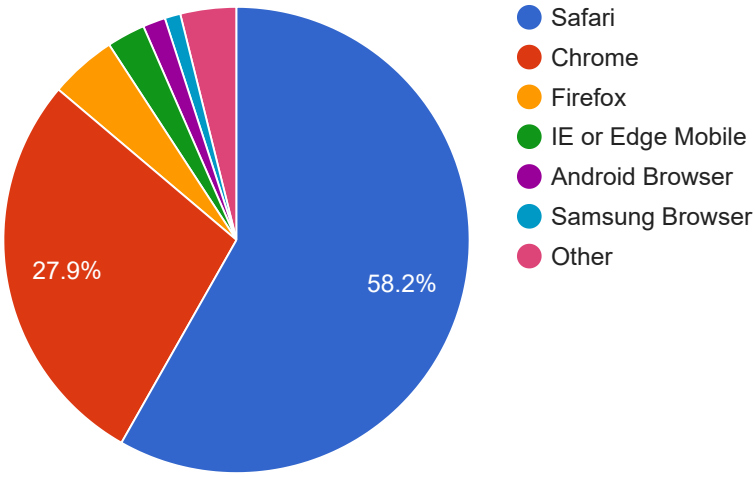
Mobile Screen Readers Used



Which of the following mobile/tablet screen readers do you commonly use?

Response	% of respondents
VoiceOver	70.6%
TalkBack	34.7%
Commentary/Jieshuo	10.1%
Voice Assistant	6.0%
VoiceView	5.9%
Mobile Accessibility for Android	4.9%
Mobile Speak	1.0%
Nuance Talks	1.0%
IDEAL	0.5%
Other	4.7%

Primary Mobile Browser

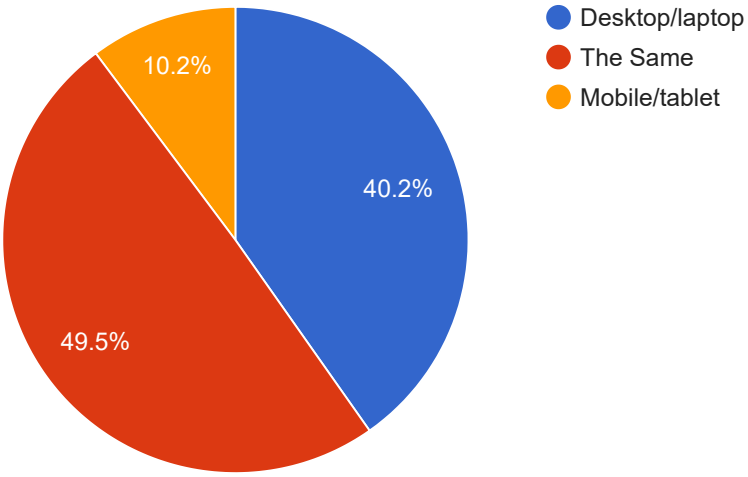


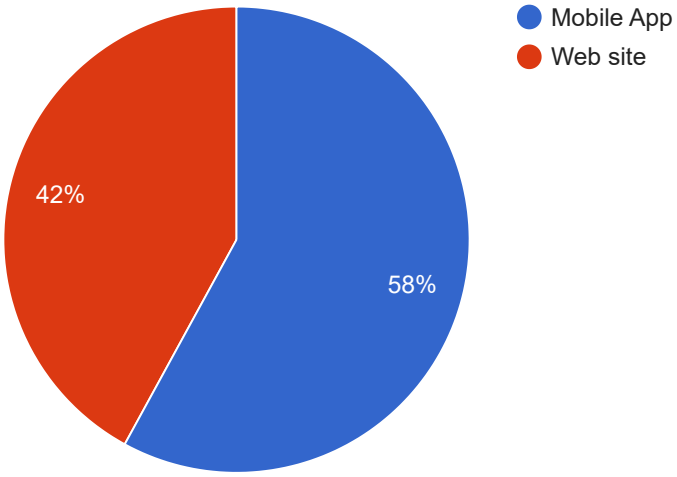
Which of the following is your primary mobile web browser?

Response	# of respondents	% of respondents
Safari	854	58.2%
Chrome	410	27.9%
Firefox	68	4.6%
IE or Edge Mobile	39	2.7%
Android Browser	23	1.6%
Samsung Browser	16	1.1%
Other	57	3.9%

Safari saw a small decrease in usage while Chrome had a slight increase in usage since 2021.

Mobile vs. Desktop/Laptop Usage



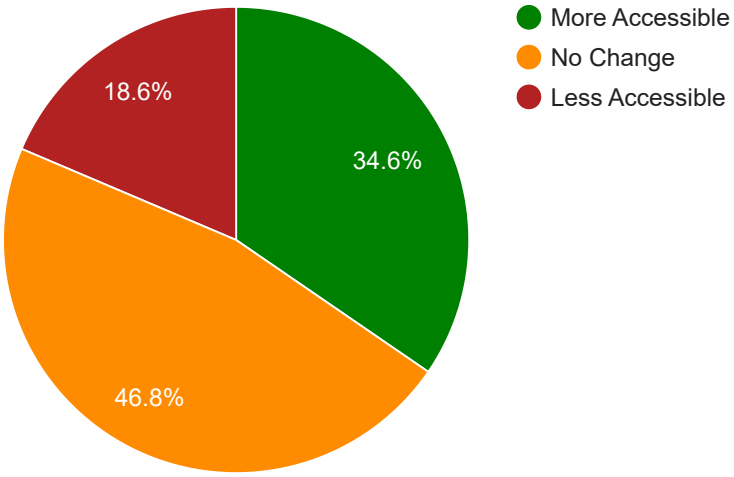


When performing common online tasks such as banking or shopping are you most likely to use a mobile app or the web site?

Response	# of respondents	% of respondents
app	852	58%
web	618	42%

Respondents indicated that they are much more likely to use a mobile app than a web site for common online tasks. The preference for mobile app usage increased to 58% in 2024, up from 51.8% in 2021 and 46% in 2017.

Web Accessibility Progress

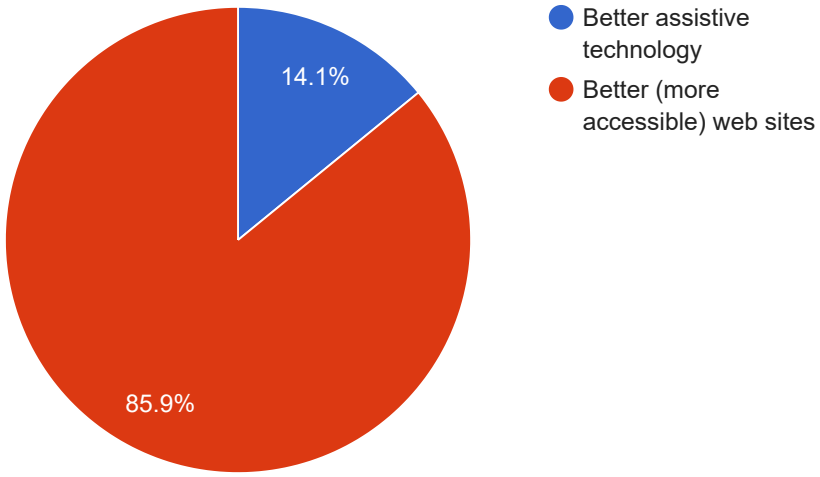


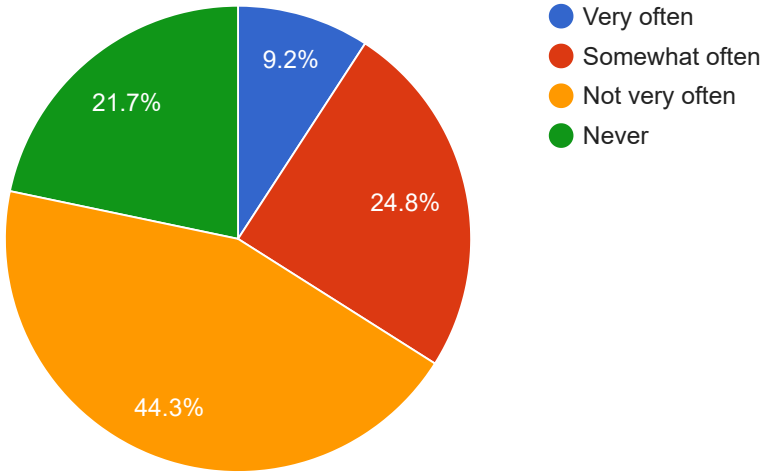
Which of the following best describes your feelings regarding the accessibility of web content over the previous year?

Response	# of respondents	% of respondents
Web content has become more accessible	522	34.6%
Web content accessibility has not changed	707	46.8%
Web content has become less accessible	281	18.6%

Perception of the state of web accessibility decreased slightly since 2021. Respondents without disabilities tend to be more positive about recent progress (45.9% thought it has become more accessible) than those with disabilities (33.4% thought it has become more accessible).

Impacts on Accessibility





How often do you contact a website owner about an accessibility barrier that you encountered?

Response	# of respondents	% of respondents
Very often	137	9.2%
Somewhat often	370	24.8%
Not very often	661	44.3%
Never	324	21.7%

The majority (67%) of respondents never or rarely contact web site owners about barriers. Respondents without disabilities reported nearly the same likelihood to contact website owners as respondents with disabilities.

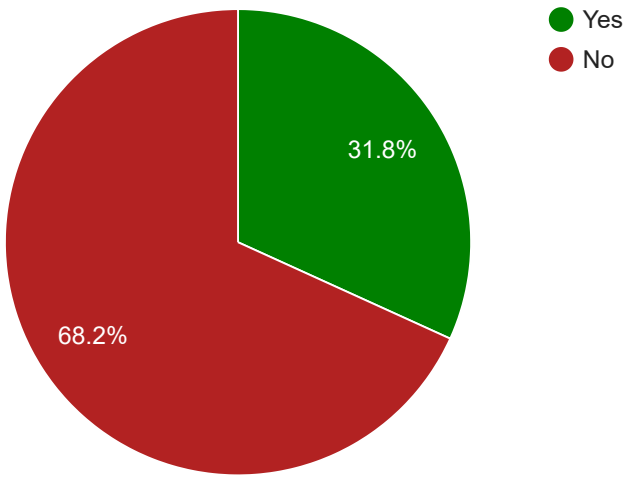
Social Media Accessibility

In general, how accessible are social media web sites to you?

Response	# of respondents	% of respondents
Very accessible	173	11.8%
Somewhat accessible	824	56.3%
Not very accessible	271	18.5%
Not at all accessible	25	1.7%
I don't know	171	11.7%

Perceptions of social media accessibility are generally unchanged over the last several years.

Descriptions in Virtual Meetings

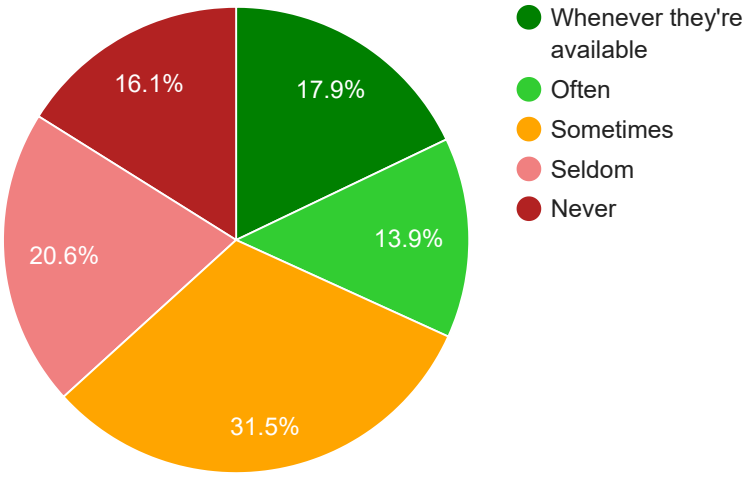


Should a person describe what they look like during a virtual meeting or webinar?

Response	# of respondents	% of respondents
Yes	363	31.8%
No	779	68.2%

The majority (68.2%) of respondents do not prefer descriptions of appearances in online meetings.

Landmarks/Regions

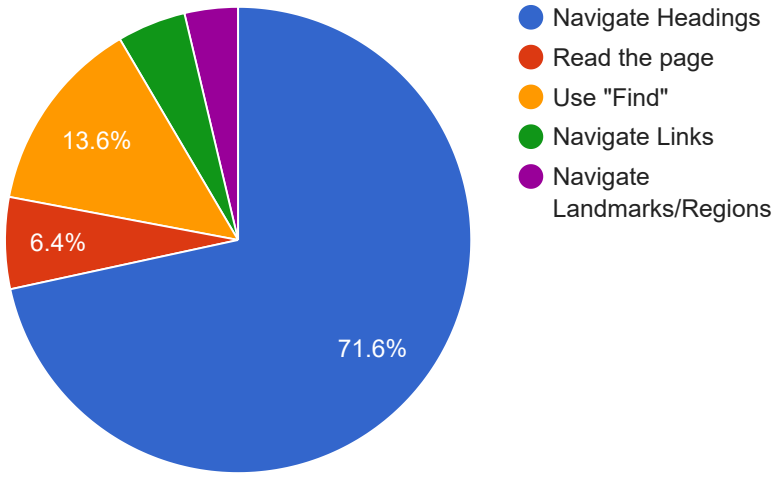


How often do you navigate by landmarks/regions in your screen reader?

Response	# of respondents	% of respondents
Whenever they're available	268	17.9%
Often	208	13.9%
Sometimes	471	31.5%
Seldom	309	20.6%
Never	241	16.1%

After steady decreases in the frequent usage of landmarks/regions was seen from 2014 (43.8%) to 2021 (25.6%). In 2024 the reported frequent usage has now increased to 31.8%.

Finding Information



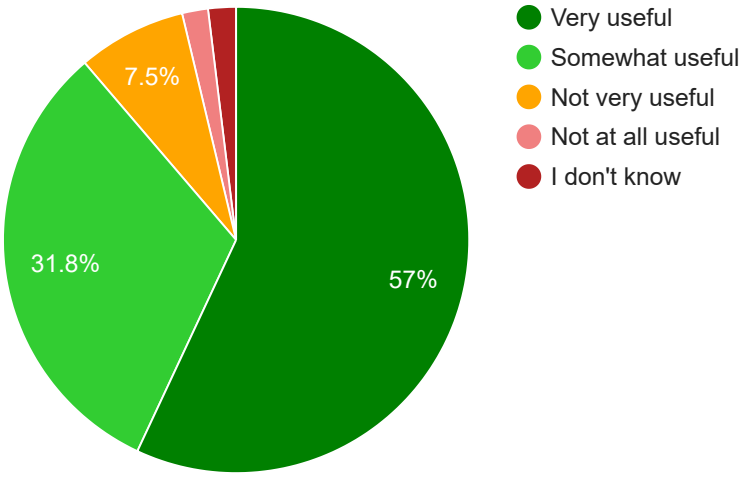
When trying to find information on a lengthy web page, which of the following are you most likely to do first?

Response	# of respondents	% of respondents
Navigate through the headings on the page	1082	71.6%
Read through the page	96	6.4%
Use the Find feature	205	13.6%
Navigate through the links of the page	72	4.8%
Navigate through the landmarks/regions of the page	56	3.7%

The usage of headings for finding information has increased over time, and it remains the predominant method. Those with advanced screen reader proficiency are much more likely to use headings (78%) than those with beginner proficiency (47%). Beginners are 2.5 times more likely to use the "Find" feature than those with advanced proficiency.

While 31.7% of respondents indicate that they always or often use landmarks when they are present, only 3.7% use this as a primary method for finding information on a lengthy web page.

Heading Levels



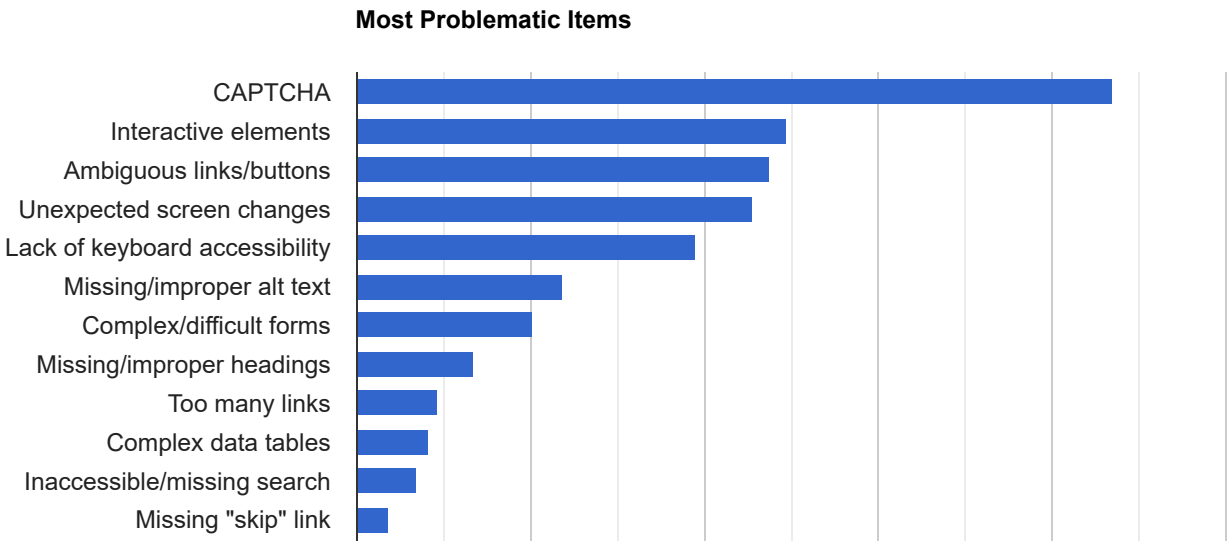
When navigating a web page by headings, how useful are the heading levels (e.g., "Heading 1", "Heading 2", etc.) to you?

Response	# of respondents	% of respondents
Very useful	855	57.0%
Somewhat useful	477	31.8%
Not very useful	113	7.5%
Not at all useful	27	1.8%
I don't know	29	1.9%

The reported usefulness of heading levels for navigation has increased over time, and 88.8% of respondents find them very or somewhat useful.

Problematic Items

The survey asked respondents to select their most, second most, and third most problematic items from a list. In giving each selected item a weighting, the following chart shows the overall rating of difficulty and frustration for each item.



In order, the most problematic items are:

1. CAPTCHA - images presenting text used to verify that you are a human user
2. Interactive elements like menus, tabs, and dialogs do not behave as expected
3. Links or buttons that do not make sense
4. Screens or parts of screens that change unexpectedly
5. Lack of keyboard accessibility
6. Images with missing or improper descriptions (alt text)
7. Complex or difficult forms
8. Missing or improper headings
9. Too many links or navigation items
10. Complex data tables
11. Inaccessible or missing search functionality
12. Lack of "skip to main content" or "skip navigation" links

The order and indicated difficulty for the items in this list are largely unchanged over the last 14 years. CAPTCHA remains the most (by a notable margin) problematic item indicated by respondents. Respondents with disabilities were twice as likely to rank CAPTCHA as a problematic item than respondents without disabilities.